BECLOMETASONE

Preventer — daily
Rinse mouth after use

CHEMICAL NAME 9-chloro-11β,17,21-trihydroxy-16β-methylpregna-1,4-diene-3,20-dione 17,21-dipropionate (beclometasone dipropionate)
CLASS Inhaled corticosteroid (ICS)
TRADE NAMES Qvar · Clenil Modulite · Becotide · Beconase (nasal) · Fostair (with formoterol)

THERAPEUTIC USES / INDICATIONS

- Asthma maintenance (preventer).
- COPD (with LABA in combination products).
- Allergic rhinitis (nasal spray).

ADMINISTRATION

- **Inhaler** twice daily; spacer recommended.
- Rinse mouth and spit after each use (oral thrush, hoarseness).
- Brands not interchangeable — extra-fine Qvar is about twice as potent as Clenil.

DOSAGES

- **Adult asthma (Clenil):** start 200 mcg twice daily; severe cases 600–800 mcg/day in 2–4 doses.
- **Qvar (extra-fine):** 100–400 mcg/day in 2 doses; max 800 mcg/day.
- **Children:** start 100 mcg twice daily (Clenil), up to 400 mcg/day; always with a spacer.
- **Nasal:** 2 sprays each nostril twice daily.

PHARMACOKINETICS

- **Onset:** benefit within days; full effect 2–4 weeks.
- Prodrug activated in lung to beclometasone monopropionate; swallowed fraction inactivated by first pass.
- Excreted mainly in faeces.

PHARMACODYNAMICS

- **Glucocorticoid receptor** activation in airways → ↓ inflammatory cytokines, eosinophils, mucosal oedema, bronchial hyper-responsiveness.
- Does not give immediate bronchodilation.

SIDE & ADVERSE EFFECTS

SIDE EFFECTS (COMMON)

- Oral candidiasis, hoarseness (dysphonia).
- Throat irritation, cough.
- Nasal: dryness, epistaxis.

ADVERSE EFFECTS (SERIOUS)

- High doses: adrenal suppression, ↓ growth in children, osteoporosis, cataract, glaucoma.
- Pneumonia (in COPD).
- Paradoxical bronchospasm.

CONTRAINDICATIONS & CAUTIONS

- Hypersensitivity.
- **Caution:** active/quiescent pulmonary TB, untreated fungal or viral respiratory infection.

KEY INTERACTIONS

- **Ritonavir, itraconazole** (CYP3A4 inhibitors): ↑ systemic steroid effects (less than fluticasone).

NURSING MANAGEMENT

- Check inhaler technique and adherence.
- Inspect mouth for thrush.
- Monitor growth in children on long-term ICS.
- Step down dose when controlled.

PATIENT EDUCATION

- Use every day even when well; it prevents attacks; it will not relieve a sudden attack.
- Rinse mouth and spit after each use.
- Benefits take 1–2 weeks.
- Carry steroid card on high doses.

KEY POINTS

- ICS preventer — anti-inflammatory.
- Daily use; not a reliever.
- Rinse mouth (thrush, hoarseness).
- Brands differ in potency.

SOURCES CHECKED **[463]** Clenil Modulite 100 micrograms inhaler (with Dose Indicator) (UK SmPC) **[464]** Qvar 100 Easi-Breathe (UK SmPC) **[465]** Beclomethasone Dipropionate, inhaled, (inhalation) (US label)
Full references p. 432–457